

titanic

May 3, 2025

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[1]: print('this is Task5')
```

this is Task5

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[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

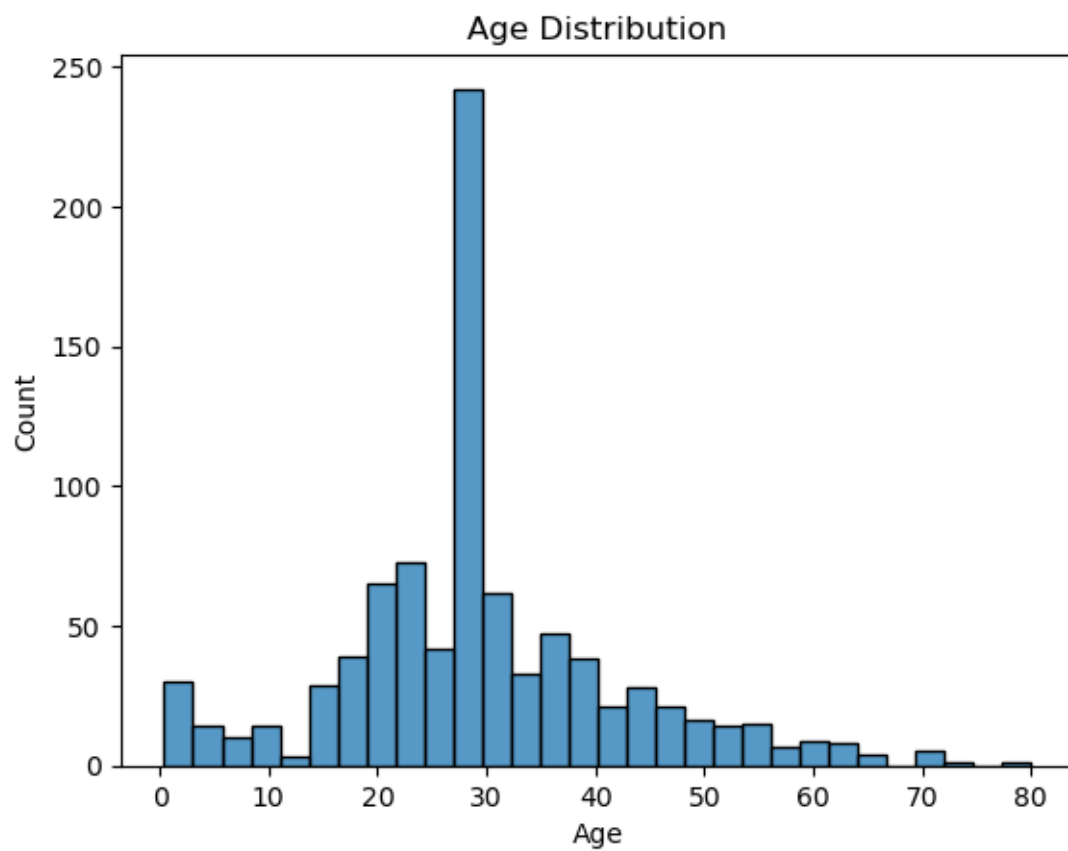
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[3]: import seaborn as sns
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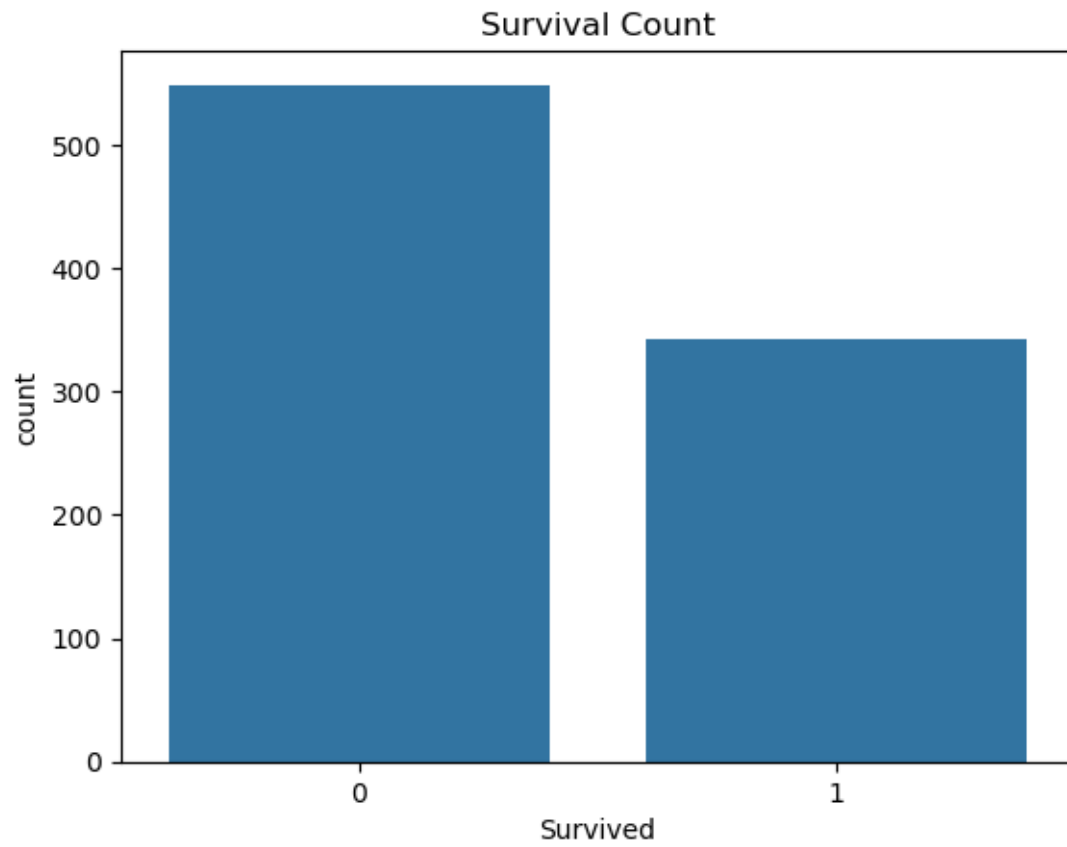
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[4]: df=pd.read_csv("titanic.csv")
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[5]: df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
if 'Cabin' in df.columns:
    df.drop(columns=['Cabin'], inplace=True)
```

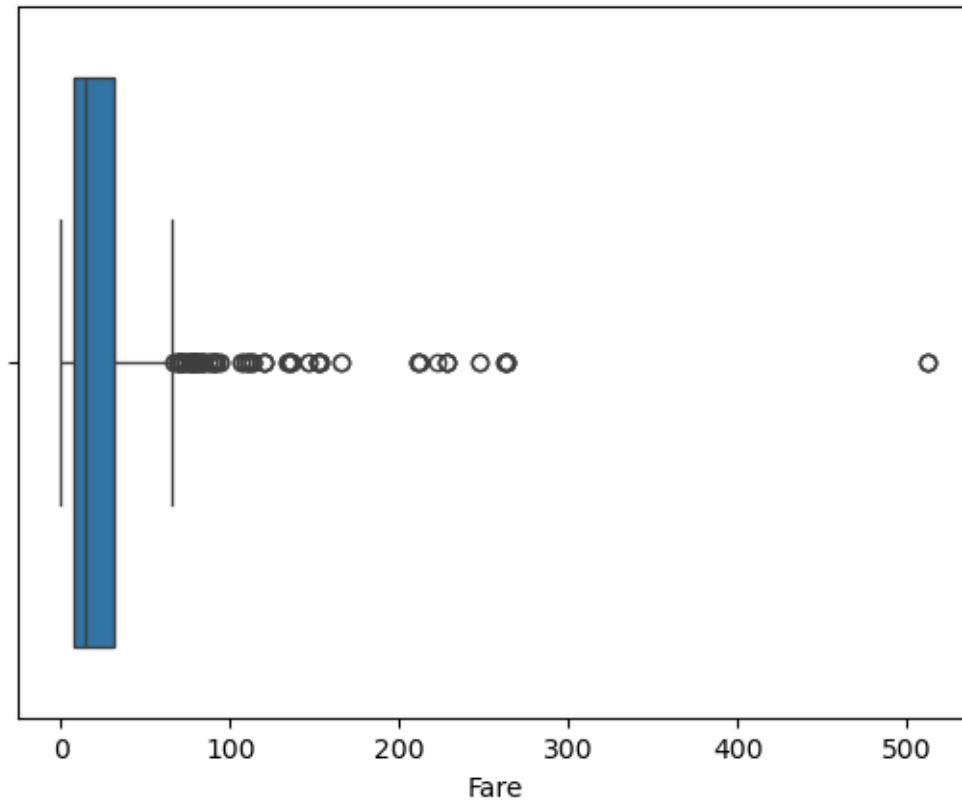
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[6]: sns.histplot(df['Age'], bins=30)
plt.title("Age Distribution")
plt.show()

sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```





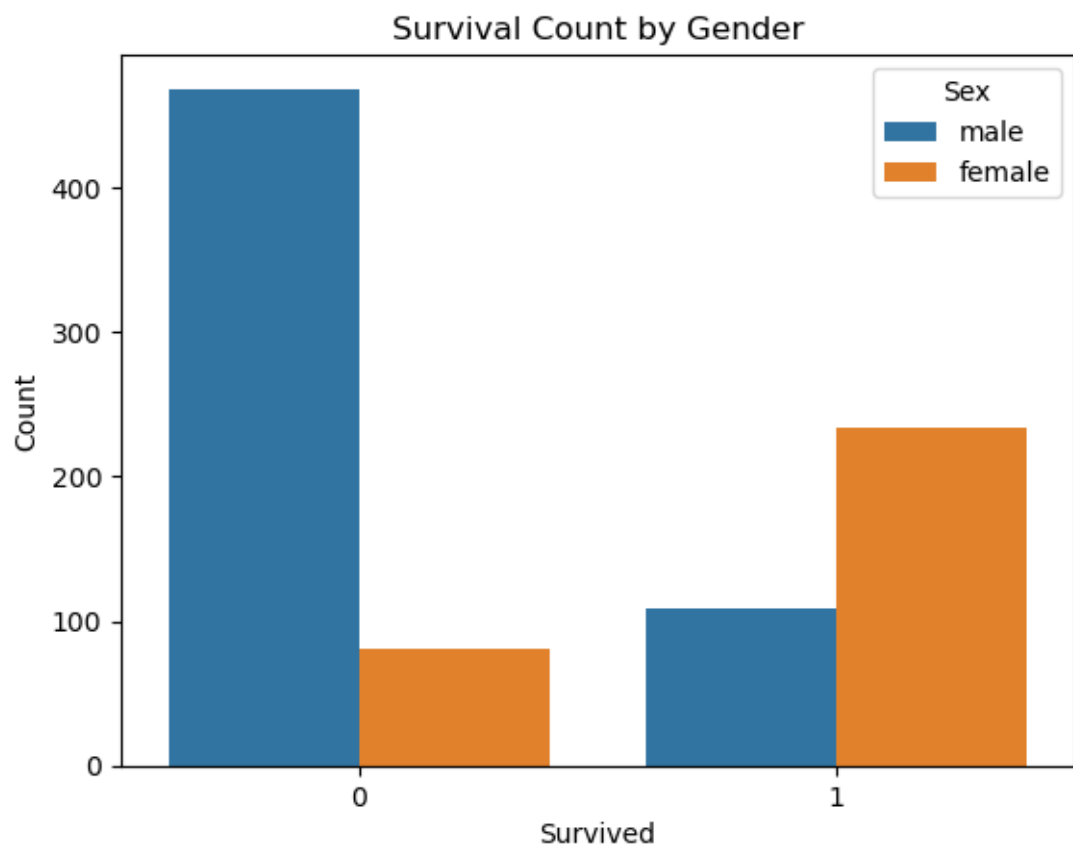
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[7]: sns.boxplot(x=df['Fare'])  
plt.show()
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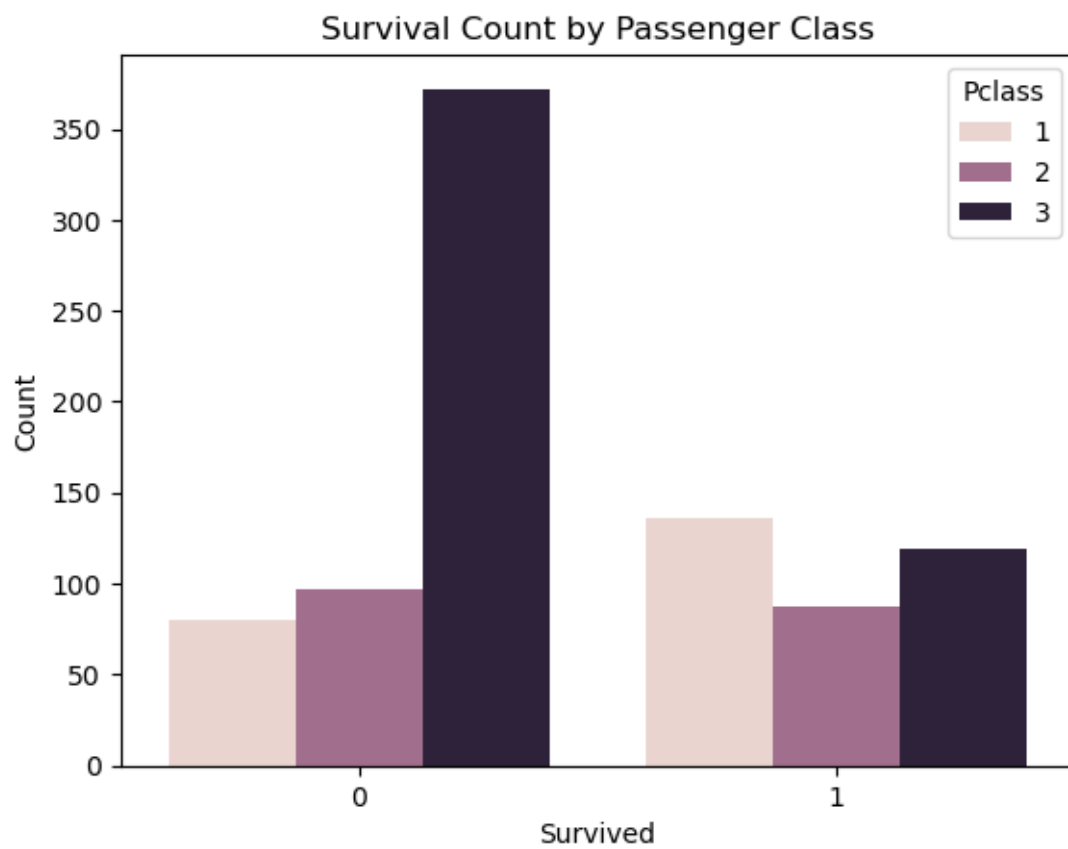


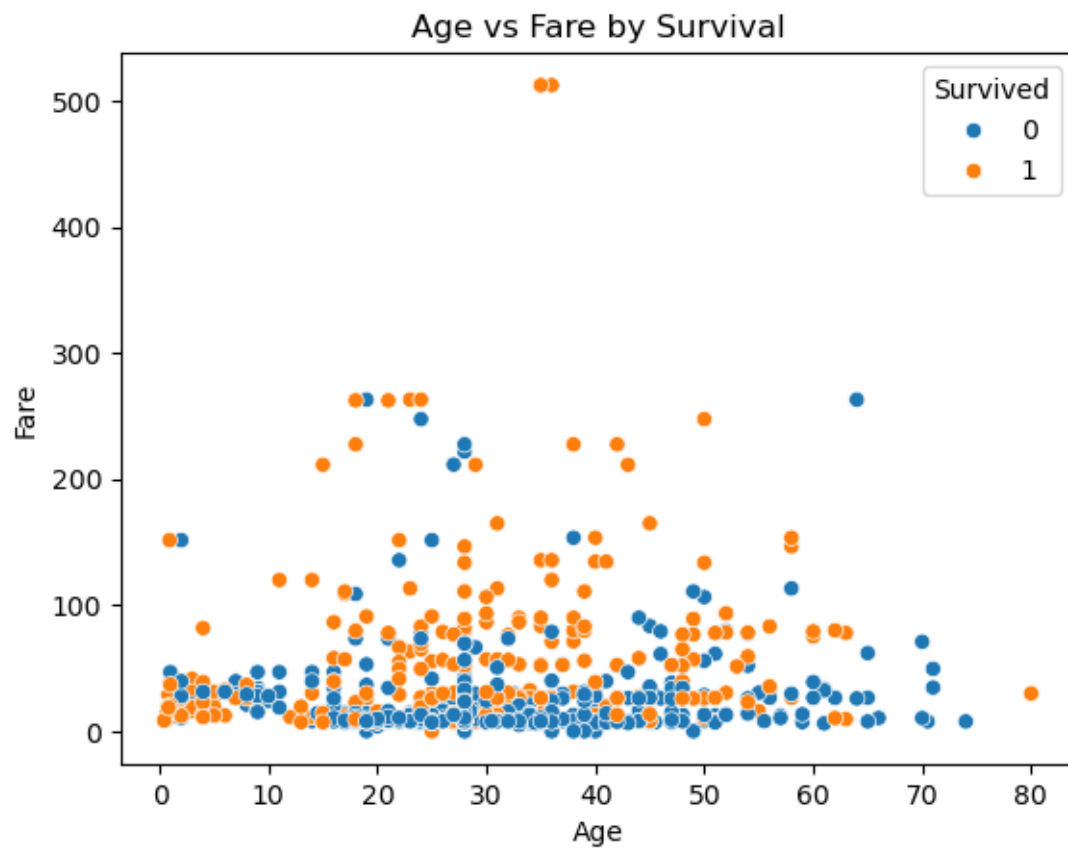
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[8]: # Survival by Sex
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title("Survival Count by Gender")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()

# Survival by Pclass
sns.countplot(x='Survived', hue='Pclass', data=df)
plt.title("Survival Count by Passenger Class")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()

# Age vs Fare
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```







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